

RELATIONSHIP BETWEEN ARTIFICIAL INTELLIGENCE (AI) AND TEACHERS' EDUCATION PROGRAMMES FOR SUSTAINABLE NATIONAL DEVELOPMENT IN ENTREPRENEURSHIP EDUCATION IN PUBLIC SCHOOLS OF NORTH CENTRAL NIGERIA

Saleh Agwom Dauda, PhD, Ogah Felix Mathew, PhD, Adepoju Oluseye Adediran

ABSTRACT

This study investigated the relationship between Artificial Intelligence (AI) and teachers' education programmes for sustainable national development in entrepreneurship education in public schools of North Central Nigeria. The study focused on how AI-driven instructional tools, intelligent tutoring systems, and adaptive learning technologies enhance teacher preparation, pedagogical competence, and innovative practices in entrepreneurship education. A descriptive survey design was adopted, involving a population of 42,316 lecturers, teacher trainers, and students from Colleges of Education and Faculties of Education offering entrepreneurship-related programmes in the North Central zone. Using Krejcie and Morgan's sampling table, a sample size of 380 respondents was selected. A structured questionnaire validated by three experts in educational management and technology achieved a Content Validity Index (CVI) of 0.82 and a Cronbach Alpha reliability coefficient of 0.81. Data were analyzed using descriptive and inferential statistics at a 0.05 significance level. The findings revealed a significant positive relationship between AI integration and teachers' education programmes in enhancing entrepreneurship skills, creativity, problem-solving, and innovation among pre-service and in-service teachers. However, the study also identified major constraints such as inadequate ICT infrastructure, poor funding, low AI literacy among teacher educators, and limited institutional support. The study concludes that the effective integration of AI into entrepreneurship teacher education is essential for achieving sustainable national development by promoting self-reliance, job creation, and technological innovation. It recommends the inclusion of AI-based entrepreneurial pedagogy in teacher training curricula and continuous professional development for educators to strengthen digital competence and instructional efficiency.

Relationship Between Artificial Intelligence (AI) And Teachers' Education Programmes For Sustainable National Development In Entrepreneurship Education In Public Schools Of North Central Nigeria

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RELATIONSHIP BETWEEN ARTIFICIAL INTELLIGENCE (AI) AND TEACHERS' EDUCATION PROGRAM

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This study investigated the relationship between Artificial Intelligence (AI) and teachers' education programmes for sustainable national development in entrepreneurship education in public schools of North Central Nigeria. The study focused on how AI-driven instructional tools, intelligent tutoring systems, and adaptive learning technologies enhance teacher preparation, pedagogical competence, and innovative practices in entrepreneurship education. A descriptive survey design was adopted, involving a population of 42,316 lecturers, teacher trainers, and students from Colleges of Education and Faculties of Education offering entrepreneurship-related programmes in the North Central zone. Using Krejcie and Morgan's sampling table, a sample size of 380 respondents was selected. A structured questionnaire validated by three experts in educational management and technology achieved a Content Validity Index (CVI) of 0.82 and a Cronbach Alpha reliability coefficient of 0.81. Data were analyzed using descriptive and inferential statistics at a 0.05 significance level. The findings revealed a significant positive relationship between AI integration and teachers' education programmes in enhancing entrepreneurship skills, creativity, problem-solving, and innovation among pre-service and in-service teachers. However, major constraints such as inadequate ICT infrastructure, poor funding, low AI literacy among teacher educators, and limited institutional support were identified. The study concludes that effective AI integration into entrepreneurship teacher education is essential for sustainable national development, promoting self-reliance, job creation, and technological innovation. It recommends the inclusion of AI-based entrepreneurial pedagogy in teacher training curricula and continuous professional development for educators to strengthen digital competence and instructional efficiency.

KEYWORDS

Artificial Intelligence, Teachers' Education Programme, Entrepreneurship Education, Sustainable National Development, Public Schools, North Central Nigeria

INTRODUCTION

Education is the cornerstone of human advancement, acting as the most potent tool for social transformation, economic growth, and sustainable national development. In the 21st century, the integration of technology-particularly Artificial Intelligence (AI)-has redefined how knowledge is created, shared, and applied. Artificial Intelligence, as defined by Russell and Norvig (2021), simulates human intelligence in machines designed for reasoning, learning, and

problem-solving...

STATEMENT OF THE PROBLEM

Teacher education is the foundation upon which the quality and sustainability of a nation's educational system are built...

RESEARCH QUESTIONS

What is the relationship between Artificial Intelligence and the National Certificate in Education (NCE) programme for sustainable national development in basic education in public schools of North Central Nigeria?

What is the relationship between Artificial Intelligence and the Professional Diploma in Education (PDE) programme for sustainable national development in basic education in public schools of North Central Nigeria?

What opportunities does Artificial Intelligence offer for improving teachers' education programmes in basic education in public schools of North Central Nigeria?

What challenges hinder the effective integration of Artificial Intelligence into teachers' education programmes for sustainable national development in basic education in public schools of North Central Nigeria?

HYPOTHESES

There is no significant relationship between Artificial Intelligence and the National Certificate in Education (NCE) programme for sustainable national development in basic education in public schools of North Central Nigeria.

There is no significant relationship between Artificial Intelligence and the Professional Diploma in Education (PDE) programme for sustainable national development in basic education in public schools of North Central Nigeria.

There is no significant relationship between Artificial Intelligence and the opportunities for improving teachers' education programmes in basic education in public schools of North Central Nigeria.

There is no significant relationship between Artificial Intelligence and the challenges that hinder effective teachers' education programmes for sustainable national development in basic education in public schools of North Central Nigeria.

METHODOLOGY

The study adopted a descriptive survey research design, considered appropriate for analyzing existing conditions and relationships without manipulating the variables under investigation...

RESULTS

RESEARCH QUESTION 1

What is the relationship between Artificial Intelligence and the National Certificate in Education (NCE) programme for sustainable national development in basic education in public schools of North Central Nigeria?

Item No	Item Description	SA	A	D	SD	X?	SD	Decision
1	AI enhances instructional delivery and pedagogical innovation in the NCE basic education programme.	168	129	42	18	3.25	0.84	Agreed
2	Integration of AI tools improves teacher trainees' digital literacy and pedagogical competence for sustainable teaching.	162	132	45	16	3.23	0.83	Agreed
3	AI supports adaptive learning systems that promote individualized instruction in NCE basic education.	159	135	44	19	3.21	0.85	Agreed
4	AI-based assessment tools enhance evaluation of students' learning outcomes in NCE programmes.	157	137	45	18	3.20	0.84	Agreed
5	AI contributes to achieving quality assurance and sustainability in basic teacher education.	160	134	43	19	3.21	0.83	Agreed
Cluster Mean		3.22	0.84	Agreed				

RESEARCH QUESTION 2

What is the relationship between Artificial Intelligence and the Professional Diploma in Education (PDE) programme for sustainable national development in basic education in public schools of North Central Nigeria?

Item No	Item Description	SA	A	D	SD	X?	SD	Decision
6	AI promotes efficient supervision and mentorship of PDE teaching practice.	163	133	41	17	3.26	0.82	Agreed
7	AI applications enhance instructional planning and content delivery for PDE students.	158	136	43	17	3.23	0.83	Agreed
8	AI supports skill-based and continuous assessment for PDE trainees.	152	139	44	19	3.19	0.84	Agreed
9	AI facilitates flexible and virtual teaching environments for PDE trainees in basic education.	155	136	45	18	3.21	0.85	Agreed
10	AI improves professional competence and readiness of PDE graduates for sustainable education.	159	133	43	17	3.23	0.84	Agreed
Cluster Mean		3.22	0.84	Agreed				

RESEARCH QUESTION 3

What opportunities does Artificial Intelligence offer for improving teachers' education programmes in basic education in public schools of North Central Nigeria?

Item No	Item Description	SA	A	D	SD	X?	SD	Decision
11	AI enhances instructional efficiency and reduces administrative workload in teacher education.	160	137	43	18	3.23	0.83	Agreed
12	AI provides real-time feedback for trainee teachers during classroom micro-teaching.	157	136	45	19	3.21	0.84	Agreed
13	AI supports the creation of digital teaching aids and lesson resources.	153	139	46	18	3.19	0.85	Agreed
14	AI promotes equal access to modern instructional resources in teacher training institutions.	155	137	44	19	3.20	0.84	Agreed
15	AI fosters continuous professional development through intelligent tutoring systems.	160	133	45	17	3.23	0.82	Agreed
Cluster Mean		3.21	0.84	Agreed				

RESEARCH QUESTION 4

What challenges hinder the effective integration of Artificial Intelligence into teachers' education programmes for sustainable national development in basic education in public schools of North Central Nigeria?

Item No	Item Description	SA	A	D	SD	X?	SD	Decision
16	Inadequate ICT and digital infrastructure limits AI integration in basic education institutions.	165	131	41	17	3.27	0.82	Agreed
17	Low AI literacy among teacher educators hinders adoption.	162	132	44	18	3.23	0.84	Agreed
18	Insufficient funding for digital innovation constrains AI adoption.	155	136	46	19	3.20	0.85	Agreed
19	Lack of clear policy and curriculum framework for AI integration in teacher education.	153	138	44	19	3.19	0.84	Agreed
20	Resistance to technological change among educators.	150	140	45	20	3.18	0.85	Agreed
Cluster Mean		3.21	0.84	Agreed				

TEST OF HYPOTHESES

TEST OF NULL HYPOTHESIS 1 (H₀?)

There is no significant relationship between Artificial Intelligence and the National Certificate in Education (NCE) programme for sustainable national development in basic education.

Responses	Observed Freq	Expected Freq	??-cal	??-tab	Df	p-value	Alpha	Decision
Agreed	297	351	228.41	13.067	350	0.011	0.05	Reject
Disagreed	55							

Interpretation: Since $\chi^2\text{-cal} (228.41) > \chi^2\text{-tab} (13.067)$ and $p\text{-value} (0.011) < 0.05$, the null hypothesis is rejected. Hence, there is a significant relationship between Artificial Intelligence and the NCE programme for sustainable national development in basic education.

TEST OF NULL HYPOTHESIS 2 (H₀)

There is no significant relationship between Artificial Intelligence and the Professional Diploma in Education (PDE) programme for sustainable national development in basic education.

Responses	Observed Freq	Expected Freq	$\chi^2\text{-cal}$	$\chi^2\text{-tab}$	Df	p-value	Alpha	Decision
Agreed	290	351	231.65	13.067	350	0.014	0.05	Reject
Disagreed	61							

Interpretation: Because $\chi^2\text{-cal} (231.65) > \chi^2\text{-tab} (13.067)$ and $p\text{-value} (0.014) < 0.05$, the null hypothesis is rejected. Therefore, there is a significant relationship between Artificial Intelligence and the PDE programme for sustainable national development in basic education.

TEST OF NULL HYPOTHESIS 3 (H₀)

There is no significant relationship between Artificial Intelligence and opportunities for improving teachers' education programmes in basic education.

Responses	Observed Freq	Expected Freq	$\chi^2\text{-cal}$	$\chi^2\text{-tab}$	Df	p-value	Alpha	Decision
Agreed	298	351	233.52	13.067	350	0.010	0.05	Reject
Disagreed	53							

Interpretation: The $\chi^2\text{-calculated}$ value (233.52) is greater than the table value (13.067) and the $p\text{-value}$ (0.010) is less than 0.05. Therefore, the null hypothesis is rejected, confirming that AI offers significant opportunities for improving teachers' education programmes in basic education.

TEST OF NULL HYPOTHESIS 4 (H₀)

There is no significant relationship between Artificial Intelligence and the challenges that hinder effective teachers' education programmes for sustainable national development in basic education.

Responses	Observed Freq	Expected Freq	$\chi^2\text{-cal}$	$\chi^2\text{-tab}$	Df	p-value	Alpha	Decision
Agreed	294	351	229.84	13.067	350	0.018	0.05	Reject
Disagreed	58							

Interpretation: Since $\chi^2\text{-calculated} (229.84) > \chi^2\text{-tab} (13.067)$ and $p (0.018) < 0.05$, the null hypothesis is rejected. This indicates a significant relationship between AI integration and the challenges affecting teachers' education programmes in basic education in North Central Nigeria.

DISCUSSION OF FINDINGS

The first major finding indicated a significant relationship between Artificial Intelligence and the National Certificate in Education (NCE) programme for sustainable national development in basic education...

CONCLUSION

The study examined the relationship between Artificial Intelligence (AI) and teachers' education programmes for

sustainable national development in basic education within public schools of North Central Nigeria...

RECOMMENDATIONS

The National Commission for Colleges of Education (NCCE) and teacher education faculties should incorporate Artificial Intelligence literacy, machine learning concepts, and digital pedagogy into NCE and PDE curricula... Regular training, workshops, and certification programmes should be organized to enhance lecturers' and teacher trainers' proficiency in AI applications...

The Federal and State Ministries of Education should increase funding allocations for ICT infrastructure in teacher training institutions...

A comprehensive policy framework should be developed to guide the integration of AI into teacher education programmes...

REFERENCES

- Adeniran, A., & Adediran, S. (2021). Technology and teacher education reform in Nigeria: Challenges and prospects. *Journal of Educational Technology*, 18(2), 56-68.
- Afe, J. O. (2020). Teacher education and national development in Nigeria: Issues and trends. *Nigerian Educational Review*, 14(1), 22-35.
- Federal Republic of Nigeria (FRN). (2014). National Policy on Education (6th ed.). Lagos: NERDC Press.
- Holmes, W., Bialik, M., & Fadel, C. (2022). Artificial intelligence in education: Promises and implications for teaching and learning. Boston: Center for Curriculum Redesign.
- Ogunode, N. J. (2023). Challenges of implementing artificial intelligence in Nigerian higher education institutions. *International Journal of Education and Information Technology*, 12(3), 41-50.
- Ololube, N. P. (2020). Teacher education and the future of Nigerian education system. *African Journal of Education Studies*, 11(4), 67-83.
- Omodara, O. D., & Adegbiya, M. V. (2022). AI integration in teacher education: Pathways to digital pedagogy. *Journal of Educational Innovation*, 15(2), 89-104.
- Oviawe, J. I. (2020). Entrepreneurship education as a tool for sustainable development in Nigeria. *International Journal of Vocational and Technical Education*, 12(1), 1-8.
- Russell, S., & Norvig, P. (2021). Artificial Intelligence: A Modern Approach (4th ed.). Pearson Education.
- Todaro, M. P., & Smith, S. C. (2020). Economic development (13th ed.). Pearson Education.
- Umar, A., & Olatunji, O. A. (2021). Professional diploma in education and quality assurance in teacher preparation in Nigeria. *Journal of Teacher Education Research*, 9(1), 24-39.
- UNESCO. (2021). AI and the futures of learning: Policy insights and recommendations. Paris: UNESCO Publishing.
- Zhang, L., & Lu, Y. (2022). Artificial intelligence and teacher education: Emerging trends and implications. *International Journal of Educational Technology*, 20(3), 99-118.